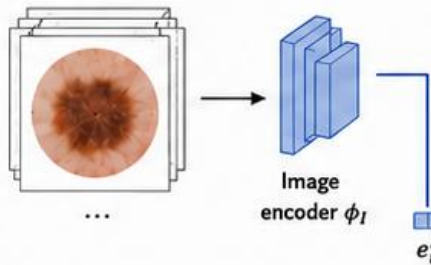
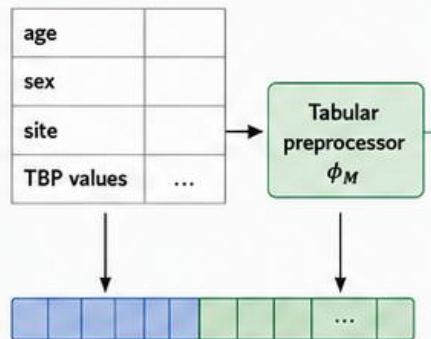


1. Multimodal Lesion Node Encoding

Cropped lesion image I_i



Metadata m_i



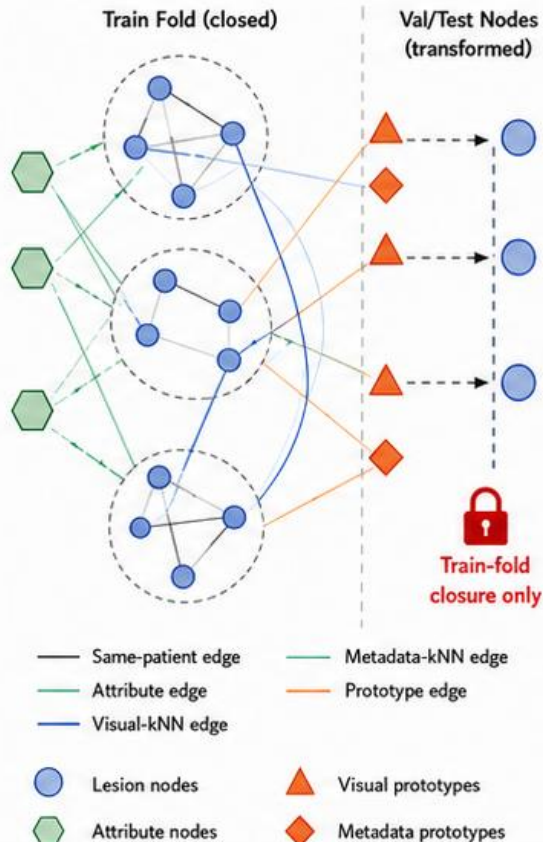
$$x_i^l = [e_{ii}^l; e_{Mi}]$$

Create lesion node features

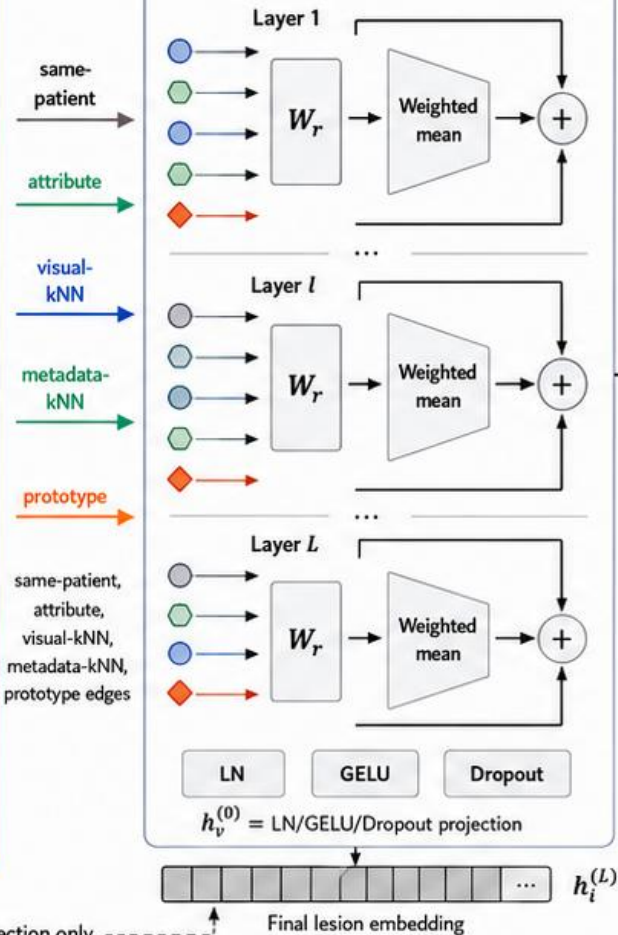
Discretized attributes: age / sex / site / size

Train-fold clustering centers

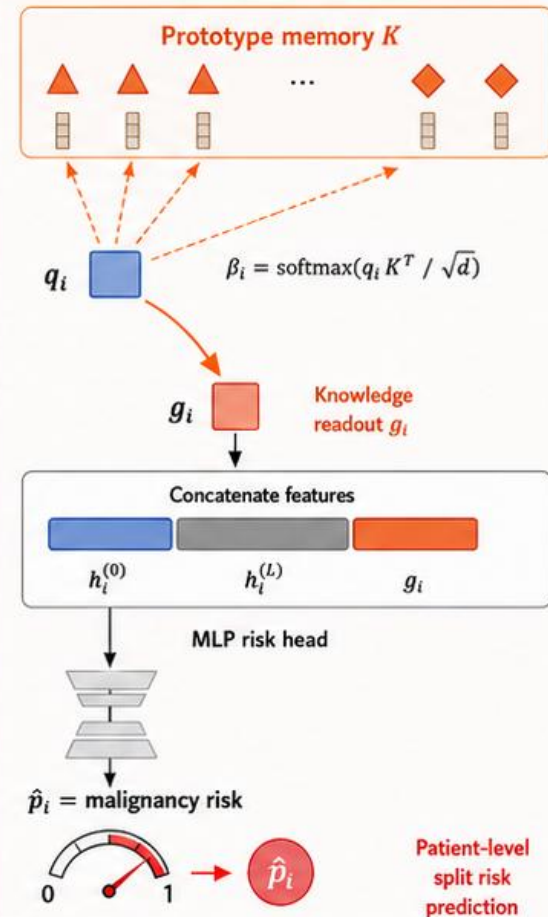
2. Fold-specific Heterogeneous Lesion Graph



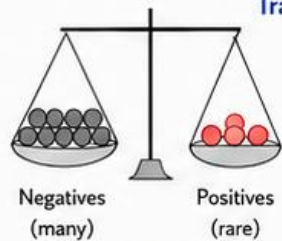
3. LGKE-GNN Relation-aware Message Passing



4. Prototype Attention & Risk Head



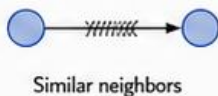
Train



$$\omega = \frac{N_{neg}}{N_{pos}}$$

Optional similarity smoothing OFF

$\lambda = 0$ OFF



Model selection only

Validation

pAUC@TPR ≥ 0.80

0.6 0.7 0.8 0.9 1.0

Validation selects checkpoint + threshold



Test



Test never tunes model

τ (fixed)

Supervised loss on train fold